

Overview of the Research Process

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Overview

- Stepwise discussion of a Research Process
- Structure of a Research Article (IMRAD Format)
- Exploring a Research Article

Steps of the Research Process

1. Identifying a Research Topic
2. Conducting a Literature Review
3. Formulating a Research Question or Hypothesis
4. Choosing the Research Methodology
5. IRB approval
6. Data Collection
7. Data Analysis
8. Interpreting Results and Drawing Conclusions
9. Writing the Research Paper
10. Citing Sources and Referencing
11. Review and Revision

Identifying a Research Topic

- Select a topic relevant to dentistry.
- Focus on a current issue or gap in the field.
- Narrow the topic to ensure it's researchable.

Example Topics:

- The role of fluoride in preventing dental caries.
- Effectiveness of teeth whitening treatments.

Conducting a Literature Review

- Search academic databases (PubMed, Google Scholar).
- Review articles, books, and previous research.
- Identify research gaps.
- **Objective:** Establish the context for your study and justify your research question.

Formulating a Research Question or Hypothesis

- Create a focused, clear, and researchable question.
 - Example: "How effective are fluoride varnishes in preventing cavities in children?"
- Develop a hypothesis based on existing knowledge.
 - Example Hypothesis: "Fluoride varnish reduces the occurrence of dental caries in children."

Choosing the Research Methodology

- **Qualitative Research:** Explores phenomena, behaviors, or attitudes (e.g., interviews, case studies).
- **Quantitative Research:** Measures variables and uses statistical analysis (e.g., surveys, experiments).
- **Mixed Methods:** Combination of both.
- **Objective:** Choose a methodology that aligns with your research question.

Institutional review Borad Approval (IRB)

- Ethical Review Panel Reviews the proposal
- The principal Investigator will defend it and then proceed for the study after it is approved.

Data Collection

- Collect data ethically, ensuring patient privacy and confidentiality.
- Use reliable and valid data collection tools (e.g., surveys, clinical trials).
- Example Tools: Questionnaires, observation, clinical measurements.

Data Analysis

- Analyze the data using statistical methods.
- Use software like SPSS or Excel for data analysis.
- Interpret the findings in relation to your hypothesis.
- **Objective:** Extract meaningful results and interpret them in the context of your research.

Interpreting Results and Drawing Conclusions

- Relate results to existing literature.
- Discuss the implications of your findings.
- Acknowledge study limitations.
- **Example:** "The fluoride varnish reduced dental caries in the experimental group compared to the control group."

Writing the Research Paper

- Structure the paper into sections:
- **Title:** Clear and concise.
- **Abstract:** Summary of the research.
- **Introduction:** Background and research question
- **Methodology:** Methods used in the research
- **Results:** Findings of the research
- **Discussion:** Interpretation of results
- **Conclusion:** Summary and future research suggestions.

Structure of a Research Article (IMRAD)

- **Title:** Concise, clear, and informative.
- **Abstract:** Summary of the key aspects of the research.
- **Introduction:** Background, research question, and objectives.
- **Literature Review:** Overview of existing research.
- **Methodology:** Research design and methods, sampling, data collection and analysis
- **Results:** Data and findings.
- **Discussion:** Analysis of results and comparisons. Implications of findings
- **Conclusion:** Key findings and recommendations.
- **References:** Citation list, source verification, academic integrity

Tips for Writing the Research Paper

- Write clearly and logically.
- Include tables, charts, and figures to present data.
- Proofread for grammar, clarity, and coherence.
- Revise the paper based on feedback from peers and mentors.
- Follow the guidelines of the target journal or institution for final submission.
- Start early and stay organized.
- Keep your research question focused.
- Use reliable and valid data sources.
- Stay ethical and respect patient confidentiality.
- Seek feedback from faculty or mentors.

Citing Sources and Referencing

- Use proper citation styles (e.g., APA, Vancouver).
- Always cite the sources you refer to in your research.
- Why it Matters: Prevent plagiarism and give credit to original researchers.

ACTIVITY: Scan the given article.

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RESEARCH ARTICLE

Open Access



Prevalence of dental caries and associated factors among primary school children: a population-based cross-sectional study in Riyadh, Saudi Arabia

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Abstract

Background: Dental caries is a preventable childhood disease, but public health efforts are hampered due to limited information on associated factors in vulnerable populations. Our study was aimed at estimating the prevalence of dental caries and identifying key associated factors in four major risk domains, including socioeconomic factors, child oral health behavior and practices, child feeding practices, and dietary habits among primary school children in Saudi Arabia.

Methods: A cross-sectional study design was used to recruit 578 male Saudi primary school children, aged 6–8 years, from 12 primary schools in five different regions of Riyadh. Children were clinically screened to detect carious lesions in primary teeth according to World Health Organization's criteria. Structured self-administered questionnaire was used to collect information on social and individual factors from the parents. The odds ratios and 95% confidence intervals of associated factors for dental caries were computed using logistic regression models; key factors were identified by systematic selection process that accounted for multicollinearity and bias correction.

Results: Dental caries was prevalent among children (83%, 95% confidence interval 79.7–86.0%). Individual factors, including irregular brushing, late adoption of brushing habit, consulting dentist for symptomatic treatment, lack of breast feeding, sleeping with a bottle in mouth, habit of snacking between meals, low consumption of fruits, and frequent consumption of soft drinks and flavored milk, were predominantly associated with dental caries in children, instead of socioeconomic factors ($p < 0.05$, adjusted R -square 80%).